

FINALTERM EXAMINATION

Fall 2009

CS301- Data Structures

Time: 120 min

Marks: 75

Student Info	
Student ID:	
Center:	
Exam Date:	01.08.2010

Question No: 1 (Marks: 1) - Please choose one

_____ only removes items in reverse order as they were entered.

► Stack

- Queue
- Both of these
- None of these

Question No: 2 (Marks: 1) - Please choose one

Here is a small function definition:

```
void f(int i, int &k)
{
    i = 1;
    k = 2;
}
```

Suppose that a main program has two integer variables x and y, which are given the value 0. Then the main program calls f(x,y); What are the values of x and y after the function f finishes?

- Both x and y are still 0.
- x is now 1, but y is still 0.
- x is still 0, but y is now 2.
- x is now 1, and y is now 2.

Question No: 3 (Marks: 1) - Please choose one

Select the one *FALSE* statement about binary trees:

▶ Every binary tree has at least one node.

- ▶ Every non-empty tree has exactly one root node.
- ▶ Every node has at most two children.
- ▶ Every non-root node has exactly one parent.

Question No: 4 (Marks: 1) - Please choose one

Every AVL is _____

- ▶ Binary Tree
- ▶ Complete Binary Tree
- ▶ None of these

▶ Binary Search Tree

Question No: 5 (Marks: 1) - Please choose one

Searching an element in an AVL tree take maximum _____ time (where n is no. of nodes in AVL tree),

- ▶ $\log_2(n+1)$
- ▶ $\log_2(n+1) - 1$

▶ $1.44 \log_2 n$

- ▶ $1.66 \log_2 n$

Question No: 6 (Marks: 1) - Please choose one

Suppose that we have implemented a *priority queue* by storing the items in a heap. We are now executing a reheapification downward and the out-of-place node has priority of 42. The node's parent has a priority of 72, the left child has priority 52 and the node's right child has priority 62. Which statement best describes the status of the reheapification.

- ▶ The reheapification is done.
- ▶ The next step will interchange the two children of the out-of-place node.
- ▶ The next step will swap the out-of-place node with its parent.
- ▶ The next step will swap the out-of-place node with its left child.

Question No: 7 (Marks: 1) - Please choose one

Suppose you implement a heap (with the largest element on top) in an array. Consider the different arrays below, determine the one that *cannot* possibly be a heap:

- ▶ 7 6 5 4 3 2 1
- ▶ 7 3 6 2 1 4 5
- ▶ 7 6 4 3 5 2 1
- ▶ 7 3 6 4 2 5 1

Question No: 8 (Marks: 1) - Please choose one

If there are 23 external nodes in a binary tree then what will be the no. of internal nodes in this binary tree?

- ▶ 23
- ▶ 24
- ▶ 21
- ▶ 22

Lesson # 27(the number of internal nodes is N, the number of external nodes will be N+1.)

Question No: 9 (Marks: 1) - Please choose one

If there are N external nodes in a binary tree then what will be the no. of internal nodes in this binary tree?

▶ N-1

▶ N+1

▶ N+2

▶ N

Question No: 10 (Marks: 1) - Please choose one

Which one of the following is NOT the property of equivalence relation:

▶ Reflexive

▶ Symmetric

▶ Transitive

▶ Associative (lesson no 34)

Question No: 11 (Marks: 1) - Please choose one

The definition of Transitivity property is

▶ For all element x member of S, $x R x$

▶ For all elements x and y, $x R y$ if and only if $y R x$

▶ For all elements x, y and z, if $x R y$ and $y R z$ then $x R z$ (lesson no 34)

▶ For all elements w, x, y and z, if $x R y$ and $w R z$ then $x R z$

Question No: 12 (Marks: 1) - Please choose one

Union is a _____ time operation.

▶ **Constant** (lesson # 35 page 11)

- ▶ Polynomial
- ▶ Exponential
- ▶ None of the given options

Question No: 13 (Marks: 1) - Please choose one

Which of the following is NOT a correct statement about Table ADT.

- ▶ In a table, the type of information in columns may be different. yes

▶ **A table consists of several columns, known as entities.** (Lesson # 38 page 1)

- ▶ The row of a table is called a record.

▶ A major use of table is in databases where we build and use tables for keeping information.

Correct A table consists of several columns, known as fields.

Question No: 14 (Marks: 1) - Please choose one

In the worst case of deletion in AVL tree requires _____.

- ▶ Only one rotation
- ▶ Rotation at each non-leaf node
- ▶ Rotation at each leaf node

▶ **Rotations equal to $\log_2 N$** (lesson # 23)

Question No: 15 (Marks: 1) - Please choose one

Binary Search is an algorithm of searching, used with the _____ data.

► **Sorted (lesson # 39)**

- Unsorted
- Heterogeneous
- Random

Question No: 16 (Marks: 1) - Please choose one

Which of the following statement is correct?

- A Threaded Binary Tree is a binary tree in which every node that does not have a left child has a THREAD (in actual sense, a link) to its INORDER successor.
- A Threaded Binary Tree is a binary tree in which every node that does not have a right child has a THREAD (in actual sense, a link) to its PREORDER successor.
- A Threaded Binary Tree is a binary tree in which every node that does not have a right child has a THREAD (in actual sense, a link) to its INORDER successor.**
- A Threaded Binary Tree is a binary tree in which every node that does not have a right child has a THREAD (in actual sense, a link) to its POSTORDER successor.

Question No: 17 (Marks: 1) - Please choose one

By using _____ we avoid the recursive method of traversing a Tree, which makes use of stacks and consumes a lot of memory and time.

- Binary tree only
- **Threaded binary tree** (lesson # 27 page 3)
- Heap data structure
- Huffman encoding

Question No: 18 (Marks: 1) - Please choose one

Which of the following statement is NOT true about threaded binary tree?

- Right thread of the right-most node points to the *dummy* node.
- Left thread of the left-most node points to the *dummy* node.

- ▶ The left pointer of dummy node points to the root node of the tree.
- ▶ Left thread of the right-most node points to the dummy node.

Lecture #28 "The left pointer of this node is pointing to the root node of the tree while the right pointer is seen pointing itself i.e. to *dummy* node. There is no problem in doing all these things. We have put the address of *dummy* node in its right pointer and pointed the left thread of the left most node towards the *dummy* node. Similarly the right thread of the right-most node is pointing to the *dummy* node. Now we have some extra pointers whose help will make the *nextInorder* routine function properly."

Question No: 19 (Marks: 1) - Please choose one

Consider a min heap, represented by the following array:

11,22,33,44,55

After inserting a node with value 66. Which of the following is the updated min heap?

▶ 11,22,33,44,55,66

▶ 11,22,33,44,66,55

▶ 11,22,33,66,44,55

▶ 11,22,66,33,44,55

Question No: 20 (Marks: 1) - Please choose one

Consider a min heap, represented by the following array:

3,4,6,7,5

After calling the function deleteMin(). Which of the following is the updated min heap?

▶ 4,6,7,5

▶ 6,7,5,4

▶ 4,5,6,7

▶ 4,6,5,7

Question No: 21 (Marks: 1) - Please choose one

We can build a heap in _____ time.

► Linear (lecture # 30 page 8)

- Exponential
- Polynomial
- None of the given options

Question No: 22 (Marks: 1) - Please choose one

Suppose we are sorting an array of eight integers using quick sort, and we have just finished the first partitioning with the array looking like this:

2 5 1 7 9 12 11 10

Which statement is correct?

- **The pivot could be either the 7 or the 9.**
- **The pivot could be the 7, but it is not the 9.**
- The pivot is not the 7, but it could be the 9
- Neither the 7 nor the 9 is the pivot.

Question No: 23 (Marks: 1) - Please choose one

Which formula is the best approximation for the depth of a heap with n nodes?

- **log (base 2) of n**
- The number of digits in n (base 10), e.g., 145 has three digits
- The square root of n
- n

Question No: 24 (Marks: 1) - Please choose one

Suppose you implement a Min heap (with the smallest element on top) in an array. Consider the different arrays below; determine the one that *cannot* possibly be a heap:

- 16, 18, 20, 22, 24, 28, 30
- 16, 20, 18, 24, 22, 30, 28
- 16, 24, 18, 28, 30, 20, 22
- **16, 24, 20, 30, 28, 18, 22**

Question No: 25 (Marks: 1) - Please choose one

While joining nodes in the building of Huffman encoding tree if there are more nodes with same

frequency, we choose the nodes _____.

► **Randomly**

- That occur first in the text message
- That are lexically smaller among others.
- That are lexically greater among others

Question No: 26 (Marks: 1) - Please choose one

Consider the following paragraph with blanks.

A is a linear list where and take place at the same end . This end is called the

What would be the correct filling the above blank positions?

- (i) queue (ii) insertion (iii) removals (iv) top
- (i) stack (ii) insertion (iii) removals (iv) bottom
- **(i) stack (ii) insertion (iii) removals (iv) top**
- (i) tree (ii) insertion (iii) removals (iv) top

Question No: 27 (Marks: 1) - Please choose one

A binary tree with 33 internal nodes has _____ links to internal nodes.

- 31
- **32** (n-1 links to internal nodes)
- 33
- 66 (2n links)

A binary tree with N internal nodes has $N+1$ external nodes.

Property: *A binary tree with N internal nodes has $2N$ links:*

$N-1$ links to internal nodes and $N+1$ links to external nodes.

Question No: 28 (Marks: 1) - Please choose one

Which traversal gives a decreasing order of elements in a heap where the max element is stored at the top?

- post-order
- level-order
- inorder

▶ None of the given options

Question No: 29 (Marks: 1) - Please choose one

What requirement is placed on an array, so that *binary search* may be used to locate an entry?

- ▶ The array elements must form a heap.
- ▶ The array must have at least 2 entries.
- ▶ The array must be sorted. (lecture # 38)
- ▶ The array's size must be a power of two.

Question No: 30 (Marks: 1) - Please choose one

Which of the following is a non linear data structure?

- ▶ Linked List
- ▶ Stack
- ▶ Queue

▶ Tree (lecture # 11 page 4)

Question No: 31 (Marks: 1)

If a Binary Tree has N internal nodes what are the no. of external nodes in it.

Sol. Lesson # 27(the number of internal nodes is N, the number of external nodes will be N+1.)

Question No: 32 (Marks: 1)

What is meant by Symmetry in equivalence relations?

Sol. Symmetry in equivalence relations mean for all elements x and y, $x R y$ if and only if $y R x$

Question No: 33 (Marks: 2)

How heap sort works to sort a set of data.

Question No: 34 (Marks: 2)

How we can apply Find operation on elements combined through Union operation.

Question No: 35 (Marks: 3)

How we can use concept of equivalence relations to generate a Maze.

Question No: 36 (Marks: 3)

Suppose we are sorting an array of eight integers using a some quadratic sorting algorithm. After four iterations of the algorithm's main loop, the array elements are ordered as shown here:

2 4 5 7 8 1 3 6

Which statement is correct? (Note: Our selectionsort picks largest items first.)

- A. The algorithm might be either selectionsort or insertionsort.
- B. The algorithm might be selectionsort, but it is not insertionsort.
- C. The algorithm is not selectionsort, but it might be insertionsort. (Correct)**
- D. The algorithm is neither selectionsort nor insertionsort.
- E. None of these.

Question No: 37 (Marks: 3)

How many leaf and non-leaf nodes are present in a complete binary tree if its depth is 7 ?

Solution:

Leaf nodes = $2^7=128$

Non-leaf nodes =127

Question No: 38 (Marks: 5)

If we insert a new element into an AVL tree of height 4, is one rotation sufficient to re-establish balance? Justify your answer.

Question No: 39 (Marks: 5)

Write down the C++ code from Selection Sort Algorithm.

Question No: 40 (Marks: 10)

Consider the following data:
the cat in the hat

- Build frequency table for the above data.
- Create a Huffman tree to determine the binary codes for each character.
- What will be the code of each letter?

a)

Character	Frequency
c	1
i	1
n	1
e	2
a	2
h	3
t	4
sp	4

c)

Character	Code
c	0000
i	0001
n	0010
e	0011
a	010
h	011
t	10
sp	11

Question No: 41 (Marks: 10)

Suppose we have build a Skip list .Now we want to add and remove items from the list .Give Algorithms for **insert (item)** and **delete (item)** methods of the Skip List.

Solution:

When we are going to insert (add) an item $(x, 0)$ into a skip list, we use a randomized algorithm. We send the item in a pair.

insert

To insert an item $(x, 0)$ into a skip list, we use a randomized algorithm:

- We repeatedly toss a coin until we get tails, and we denote with i the number of times the coin came up heads
- If $i \geq h$, we add to the skip list new lists $Sh+1, \dots, Si+1$, each containing only the

two special keys

- We search for x in the skip list and find the positions $p_0, p_1, \dots, p_i$ of the items with largest key less than x in each list $S_0, S_1, \dots, S_i$
- For $j \leftarrow 0, \dots, i$, we insert item (x, o) into list S_j after position p_j

delete

To remove an item with key x from a skip list, we proceed as follows:

- We search for x in the skip list and find the positions $p_0, p_1, \dots, p_i$ of the items with key x , where position p_j is in list S_j
- We remove positions $p_0, p_1, \dots, p_i$ from the lists $S_0, S_1, \dots, S_i$
- We remove all but one list containing only the two special keys